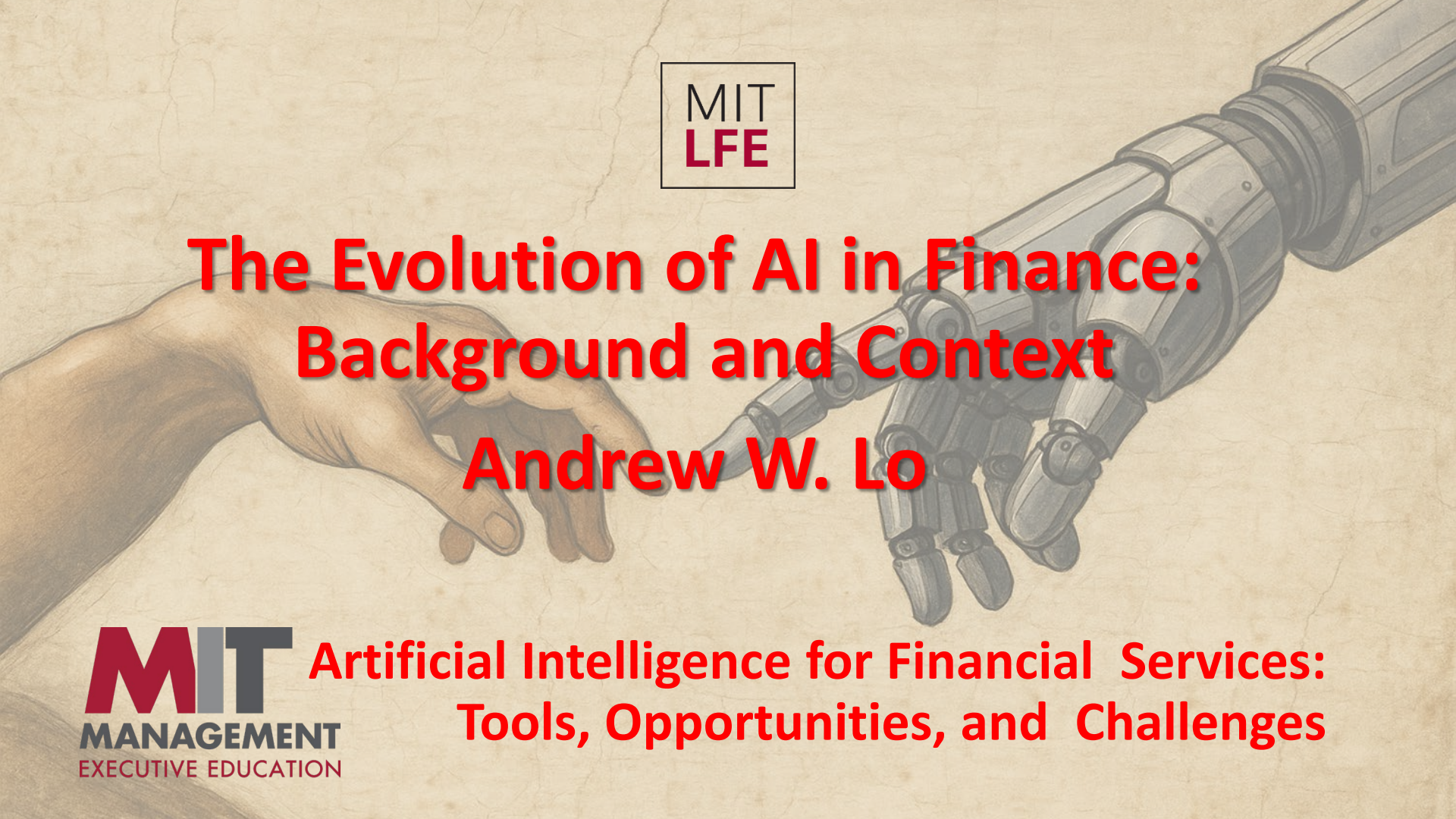




MIT
LFE

The Evolution of AI in Finance: Background and Context

Andrew W. Lo

MIT
MANAGEMENT
EXECUTIVE EDUCATION

Artificial Intelligence for Financial Services:
Tools, Opportunities, and Challenges

Outline

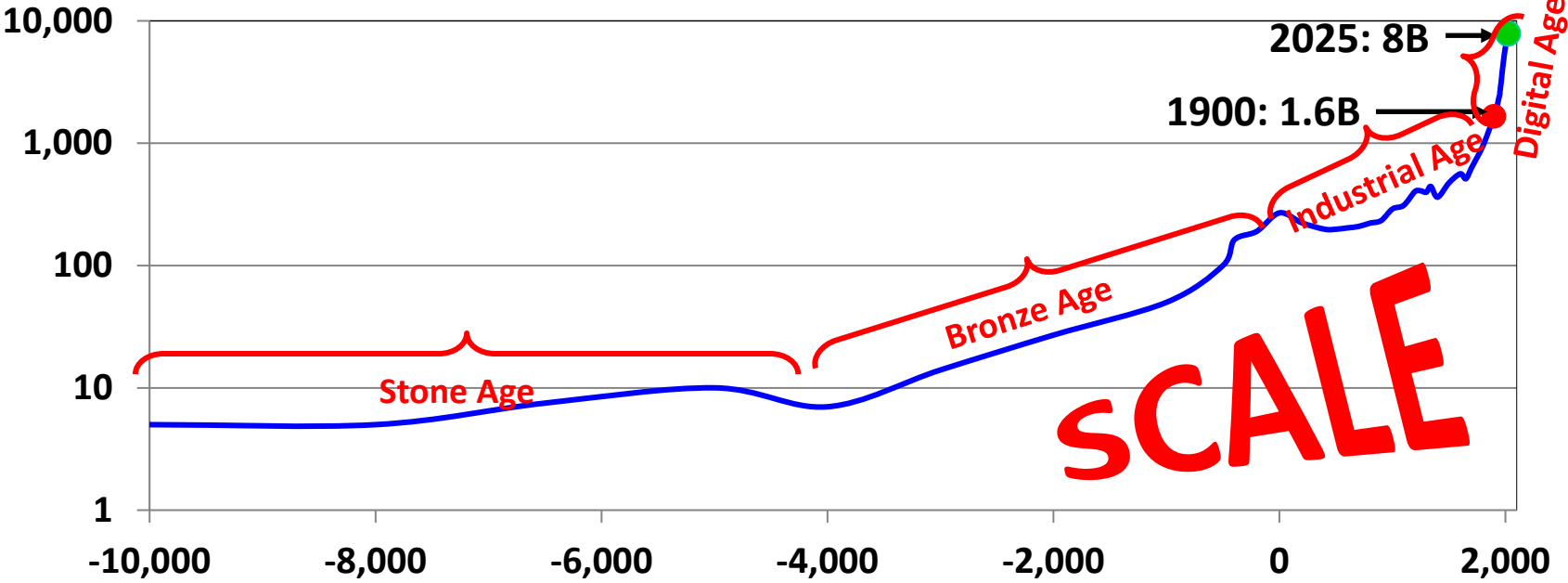
- Motivation
- Rationality
- Modeling Human and Artificial Intelligence
- A Brief History of AI
- Narrative

World Population of *Homo Sapiens* 10,000 BC to 2050 AD



Technology

World Population of *Homo Sapiens* 10,000 BC to 2050 AD



Technology

Deep Technology

Scales:

Quantum Computing	→	Computation
Drug Development	→	Lifespan
Climate Technology	→	Environment
Nuclear, Geothermal	→	Clean Energy
Robotics	→	Physical Labor
Space Travel	→	Population
Artificial Intelligence	→	Intelligence



So What Have We Learned?

**Markets
are
efficient**

Rationality



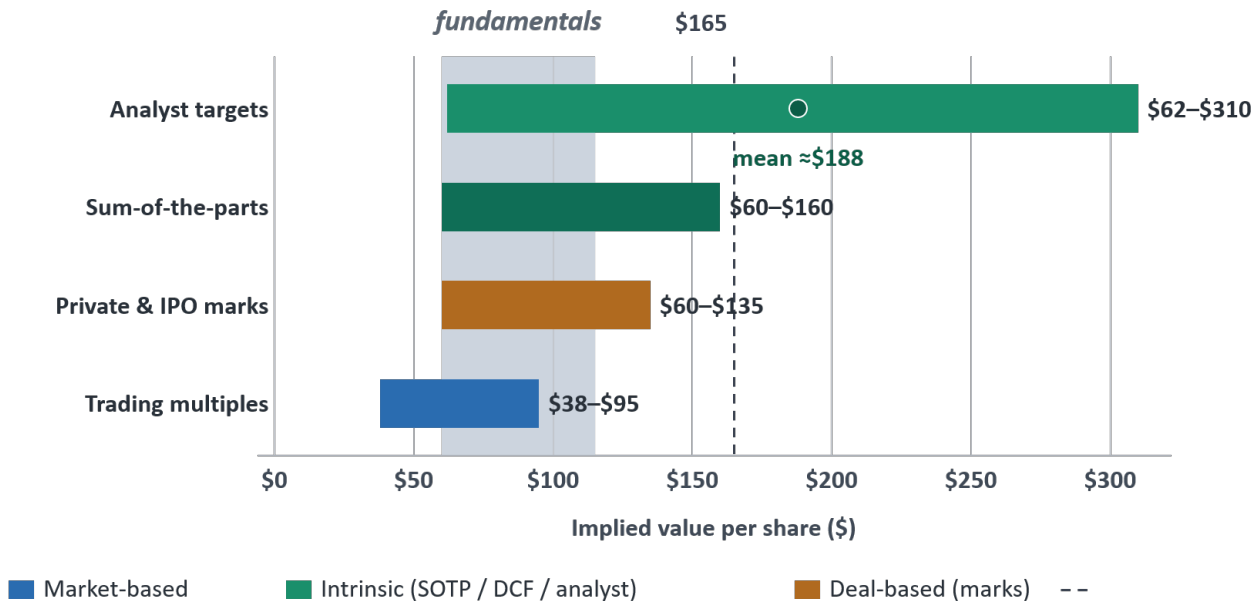
**People
behave
irrationally**

**Animal
Spirits**

Example: SpaceX

Price as of 7/22/2026: **\$115.98**

SpaceX Valuations as of 6/22/2026 (\$165)





Personal Journey

Efficient Markets
Rational Expectations



Behavioral Finance
Psychology

Artificial Intelligence
Bounded Rationality

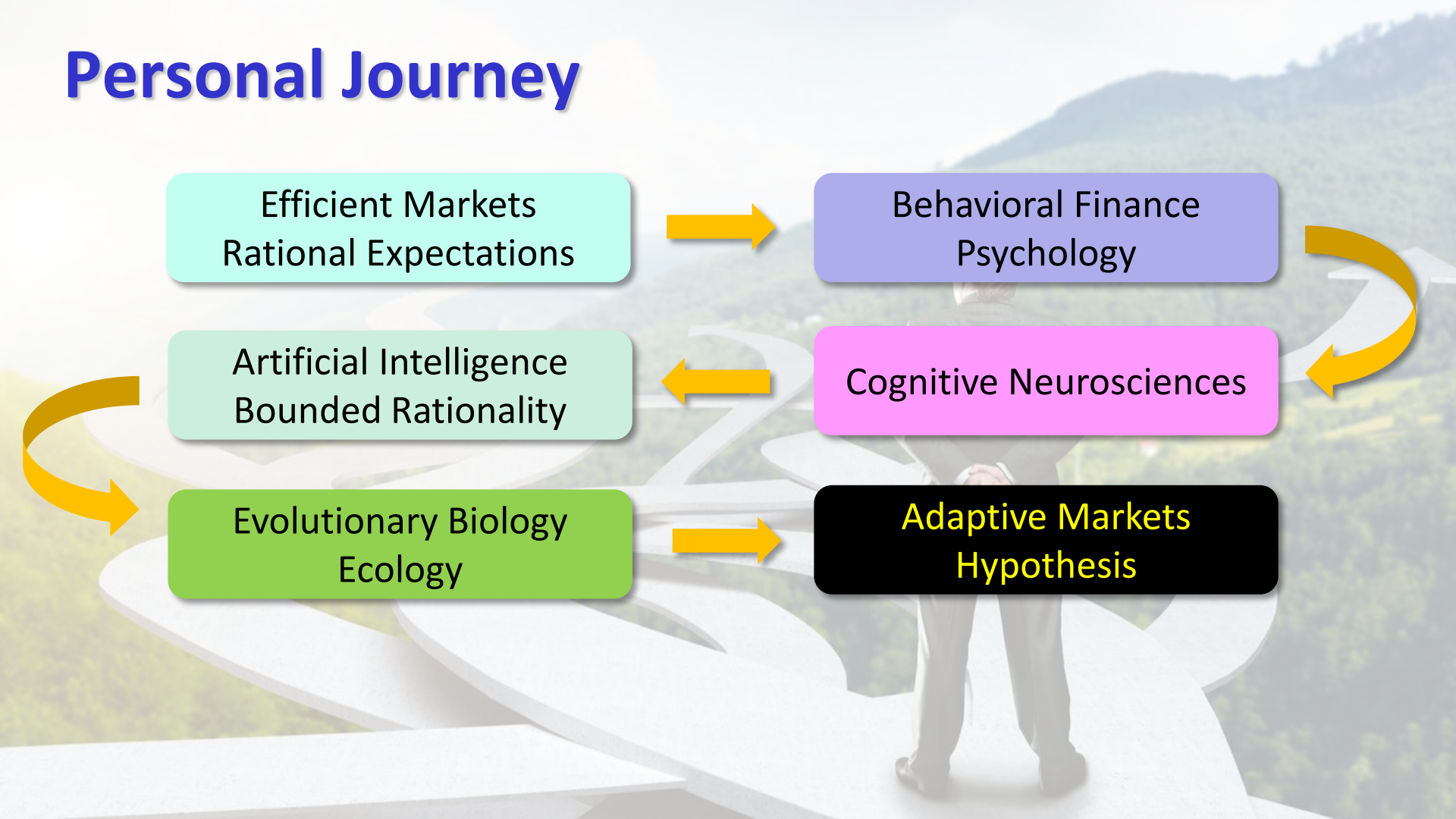


Cognitive Neurosciences

Evolutionary Biology
Ecology



Adaptive Markets
Hypothesis

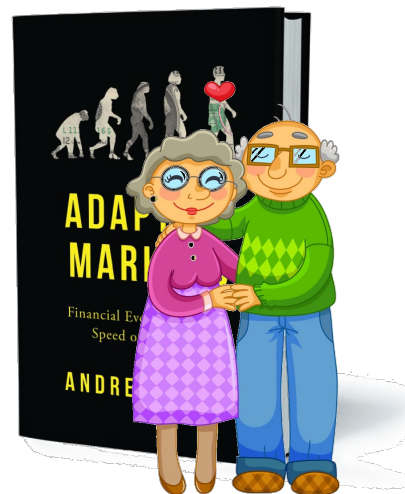


A New Finance Paradigm Is Emerging

“Nothing makes sense in biology except in the light of evolution,”
Dobzhansky (1973)

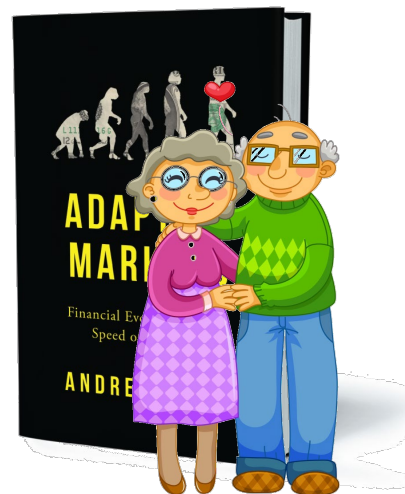
“Nothing makes sense in the financial industry except in the light
of the **Adaptive Markets Hypothesis**,” Lo (2017)

1. Individuals act in their own self-interest
2. Individuals make mistakes (“satisfice”)
3. Individuals learn and adapt (heuristics)
4. Competition drives adaptation and innovation
5. Evolution determines market dynamics



A New Finance Paradigm Is Emerging

- EMH Not wrong, merely incomplete
- Stable economy $\Rightarrow$ stable laws (EMH)
- Dynamic economy $\Rightarrow$ violations (AMH)
- The current environment is highly **dynamic**, e.g., geopolitics, war(s), and AI
- We must **adapt** to changing market conditions



Technology Has Always Impacted Finance

Wall Street Before the
Invention of the Telegraph



Wall Street After the
Invention of the Telegraph



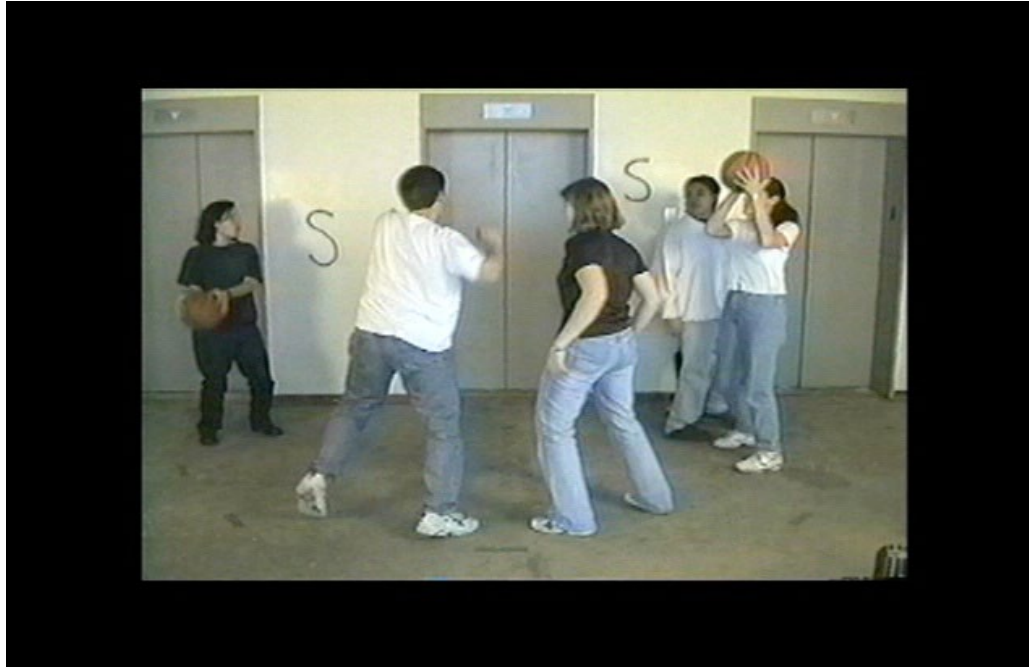
Technological
Innovation
Creates
Winners and
Losers—But
How Fast?

How (Quickly)
Can We Adapt??

Source: Leinweber, 2009, *Nerds on Wall Street*.

Rationality

Perception and Cognitive Load



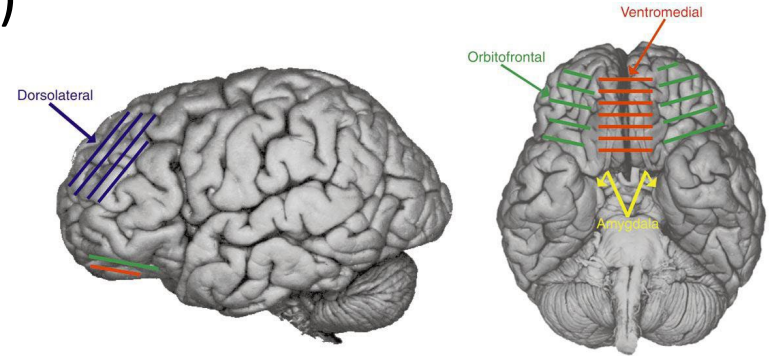
www.viscog.com

Prefrontal Cortex (PFC)

Higher Thought (Uniquely Human Behaviors)

- Planning, goal-oriented behavior, “executive function”, logical deliberation, abstract thoughts
- Serial integration of information from other components (limited capacity)
- If *Homo economicus* exists, this is the region

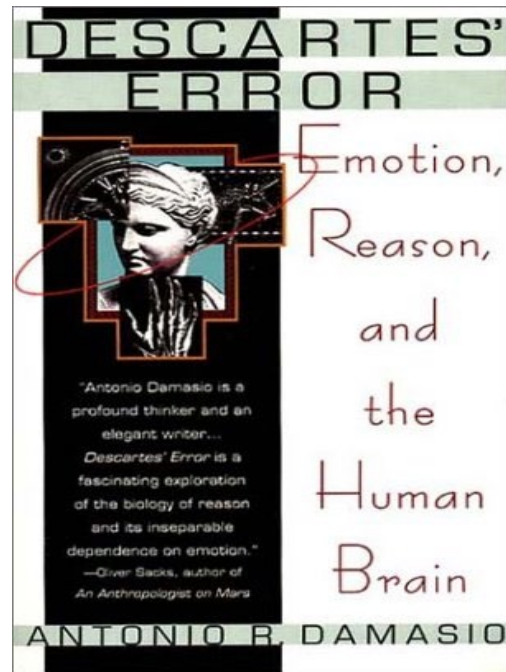
But This Isn't Quite “Rationality”



Rethinking Rationality

Patient Elliot Had Frontal Lobe Damage:

- No Impact on IQ or Logical Functions
 - Perceptual Ability
 - Past Memory
 - Short-Term Memory
 - Learning
 - Language
 - Arithmetic
- But Behavior Was Quite Irrational!



Damasio (1994)

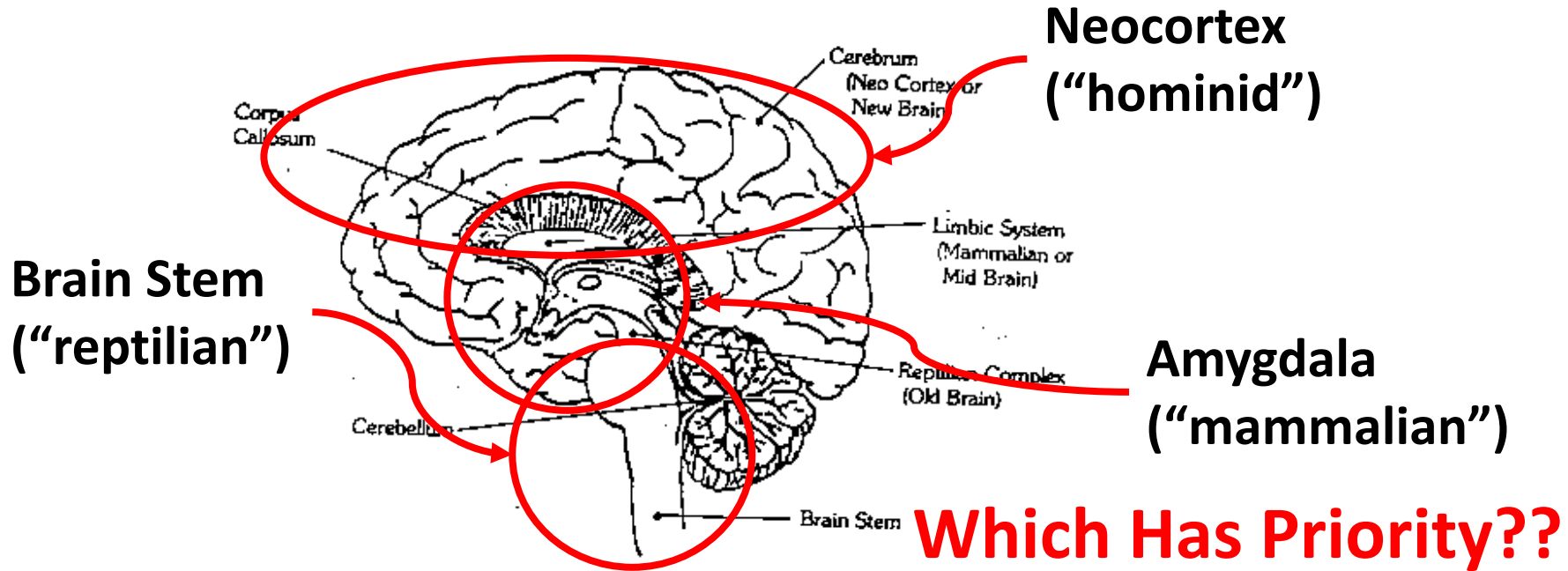
Rethinking Rationality

Damasio (1994, p. 36):

When the job called for interrupting an activity and turning to another, he might persist nonetheless, seemingly losing sight of his main goal. Or he might interrupt the activity he had engaged, to turn to something he found more captivating at that particular moment... The flow of work was stopped. One might say that the particular step of the task at which Elliot balked was actually being carried out too well, and at the expense of the overall purpose. One might say that Elliot had become irrational concerning the larger frame of behavior...

Rethinking Rationality

Multiple Components Interact To Produce Behavior



The Stroop Task

Say the colors of the following word:

RED	GREEN	BLUE	YELLOW	ORANGE
BLUE	BROWN	RED	GREEN	PURPLE
PINK	BLACK	BLUE	YELLOW	GREEN

The Stroop Task

Say the colors of the following word:

RED

GREEN

BLUE

YELLOW

ORANGE

BLUE

BROWN

RED

GREEN

PURPLE

PINK

BLACK

BLUE

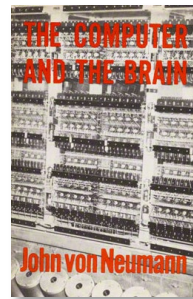
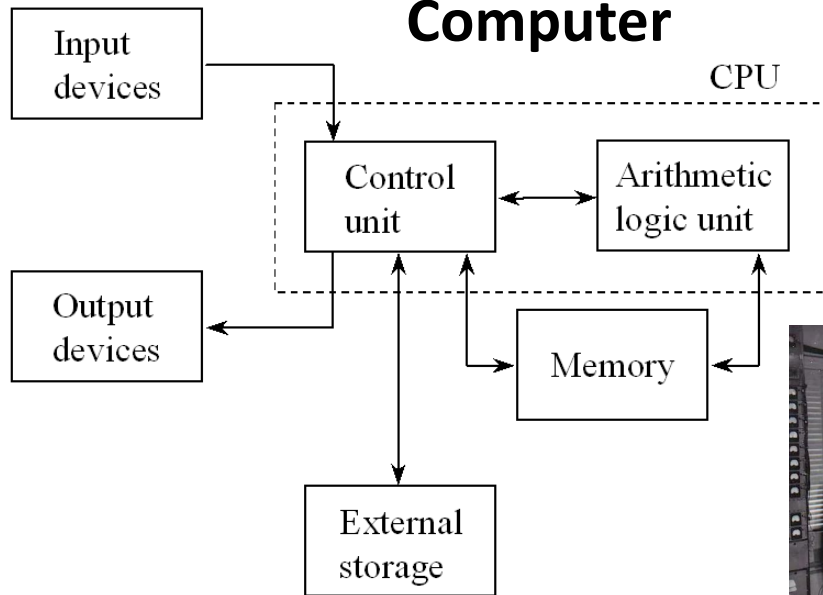
YELLOW

GREEN

Modeling Human and Artificial Intelligence

How Do Humans Learn?

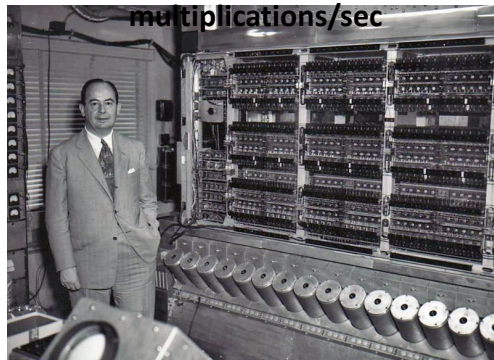
The Modern Digital Computer



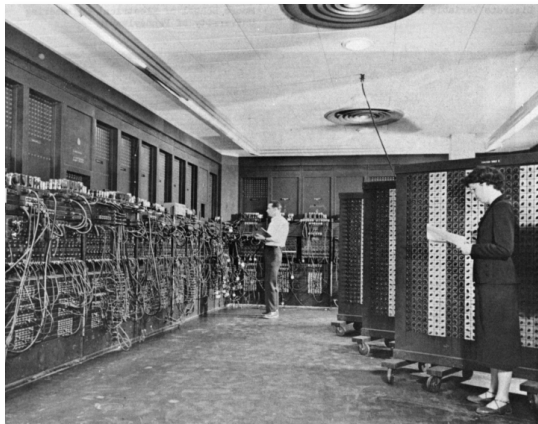
IAS Computer (1942)

140

multiplications/sec



How Do Humans Learn?



ENIAC, 1946

- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- 500 flops

How Fast Are Today's Computers?

- MFlop = 1,000,000 FP Operations / Second

LineShine (NSCS, Shenzhen)

2.198 exaflops (10^{18} flops)

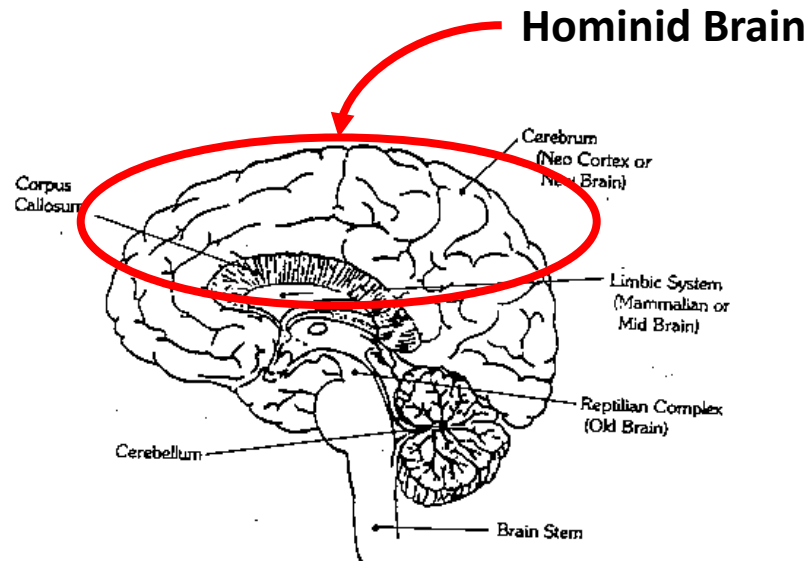
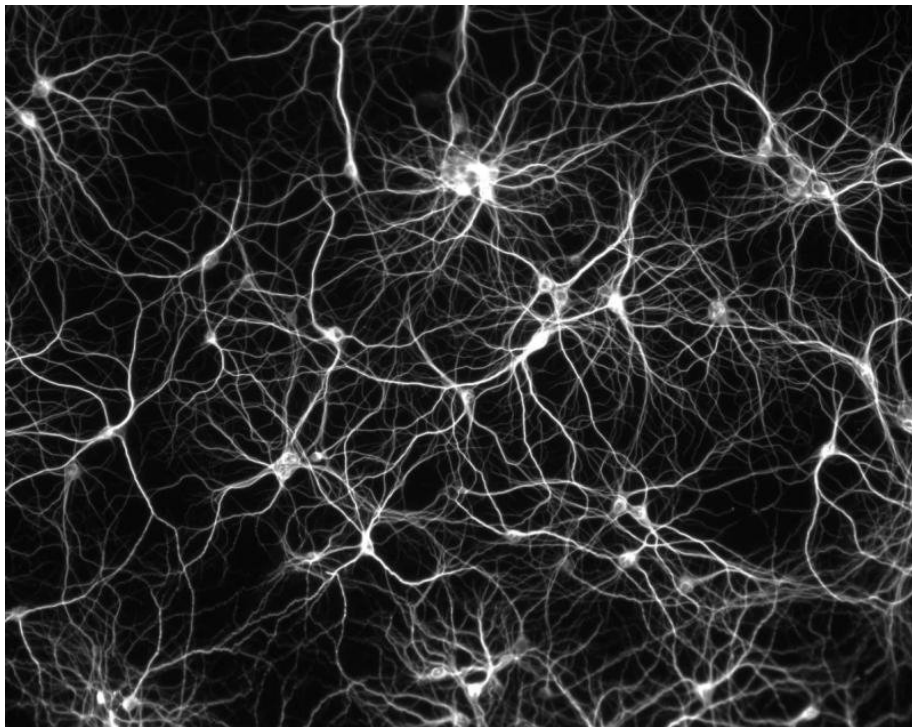
June 23, 2026

China Takes Supercomputer Crown From U.S. for First Time Since 2017

A supercomputer in Shenzhen was declared the world's fastest. It uses only standard microprocessors and not the special-purpose chips called graphics processing units.



How Do Humans Learn?



How Do Humans Learn?

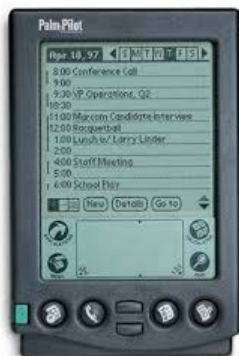
Identify Subject As Quickly As Possible

How Do Humans Learn?

Identify Emotion As Quickly As Possible

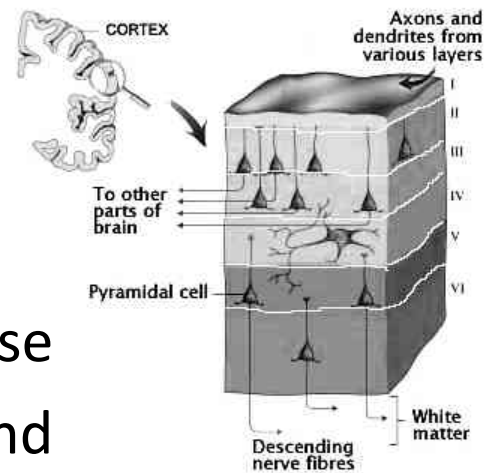
The Memory/Prediction Model

- A neuron takes 5 milliseconds to transmit an impulse
- That means a neuron can fire 200 times/second
- LineShine is ~11 quadrillion times faster (30 megawatts)
- But humans can identify faces in 0.5 second (20 watts)
- At most 100 steps for human pattern recognition
- How??
- Jeff Hawkins (2004)



The Memory/Prediction Model

- Vernon Mountcastle's (1978) discovery:
 1. The neocortex is anatomically homogeneous
 2. Six layers are organized in columns
- This implies that “software” is the key
- Hawkins' memory/prediction model
 - Neocortex is mainly a large memory warehouse
 - Intelligence consists of two things: memory and prediction



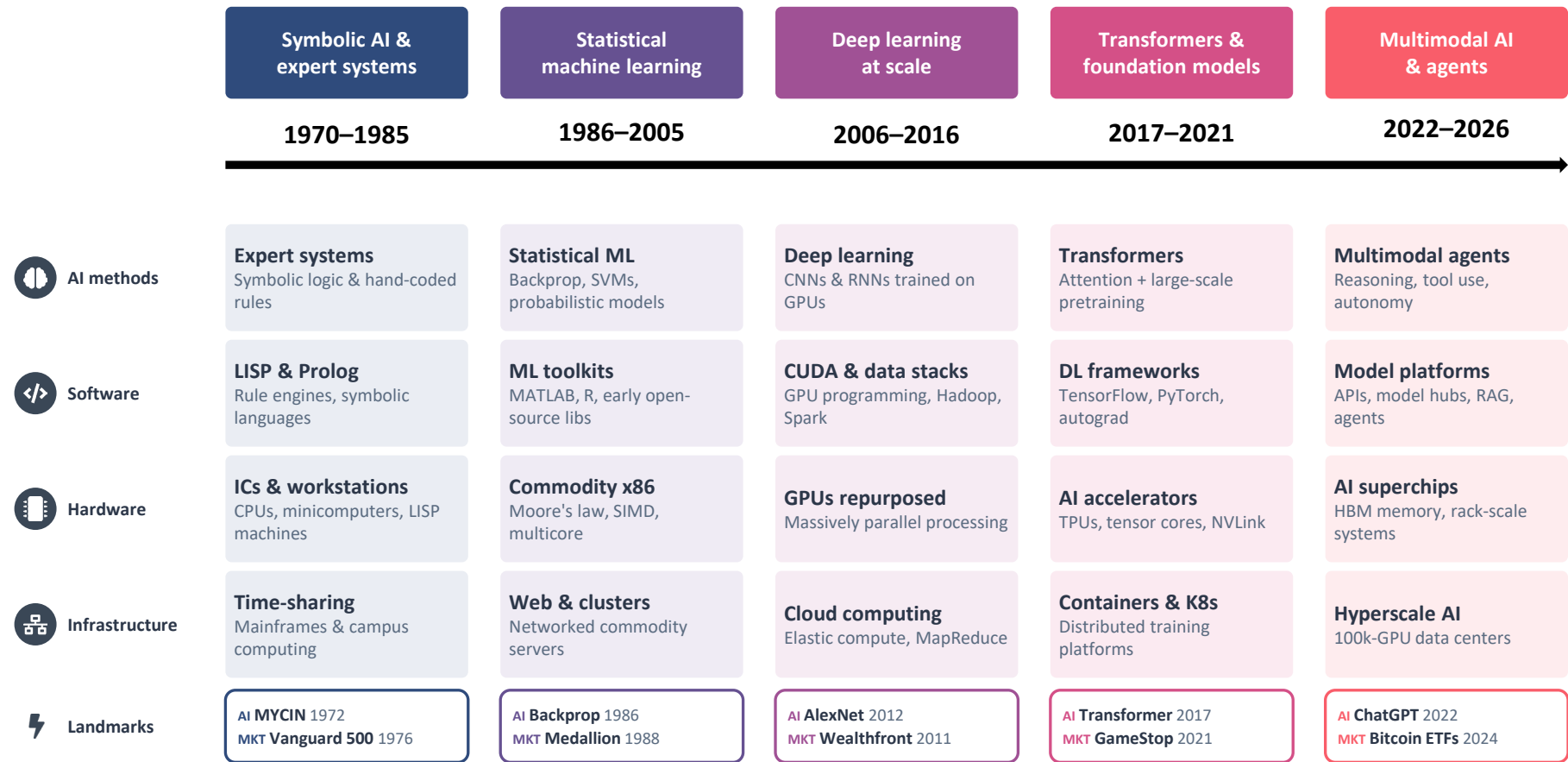
The Memory/Prediction Model

- Humans store knowledge differently than computers
 - Computers store data
 - We store sequences of events, stories, **narrative**
 - Matteo Ricci's (1596) *Treatise on Mnemonics*
- Survival depends on prediction
- Most potent fear: fear of the **unknown**
- Prediction is instinctive in humans
- Traditional computing cannot compete yet (maybe??)
- How we create, store, and retrieve **narratives** is key
- Three-year old's "why?"

A Brief History of AI

Innovation in the AI Technology Stack

1970–2026: each wave of investment innovation arrived in step with new AI methods and the computing stack beneath them



Periods mark eras of prominence, not hard boundaries. Earlier investing waves — value investing (1930s), MPT (1950s), growth investing (1960s) — predate the computing stack shown.

The First Generation: Expert Systems

1970s: **MYCIN** by Edward Shortliffe

Key Features



Medical Diagnosis

Diagnosed blood infections and recommended antibiotics for treatment



Rule-Based System

Contained ~600 production rules encoding expert medical knowledge



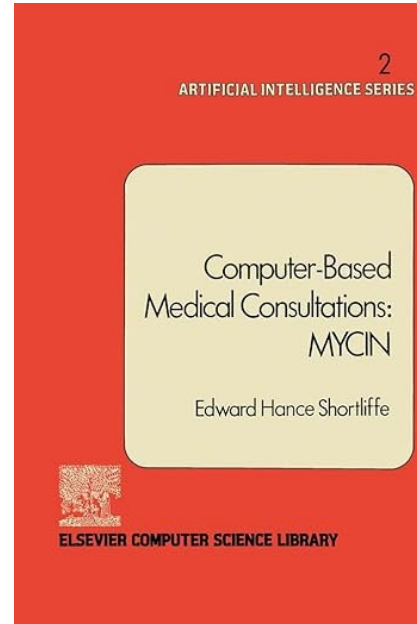
Backward Chaining

Used goal-driven inference to work backwards from hypotheses to supporting evidence



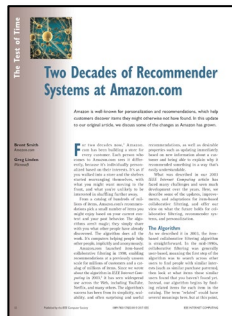
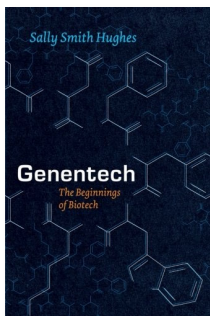
Certainty Factors

Pioneered the use of certainty factors to handle uncertain medical knowledge



Second Generation: Machine Learning

1990s:



- Emerged via advances in data, storage, and compute costs
- Expensive storage $\Rightarrow$ small data, complex code
- Cheap storage $\Rightarrow$ big data, simple code
- This is closer to natural intelligence! **Narrative** vs. facts

Mimicking Human Intelligence

msot Teh hvea fro tormnet is ot hatuefl mne
fo evreyhtnig ntoginh pwore btu oevr kwnoeldeg

Teh msot hatuefl tormnet fro mne is ot hvea
kwnoeldeg fo evreyhtnig btu pwore oevr ntoginh

Mimicking Human Intelligence

msot Teh hvea fro tormnet is ot hatuefl mne
fo evreyhtnig ntoginh pwore btu oevr kwnoeldeg

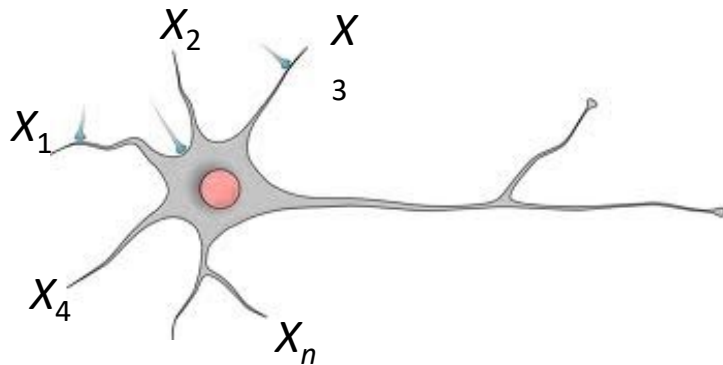
The most hateful torment for men is to have
knowledge of everything but power over nothing

– Herodotus Histories bk.9
sec.16

Third Generation: ANNs, Deep Learning

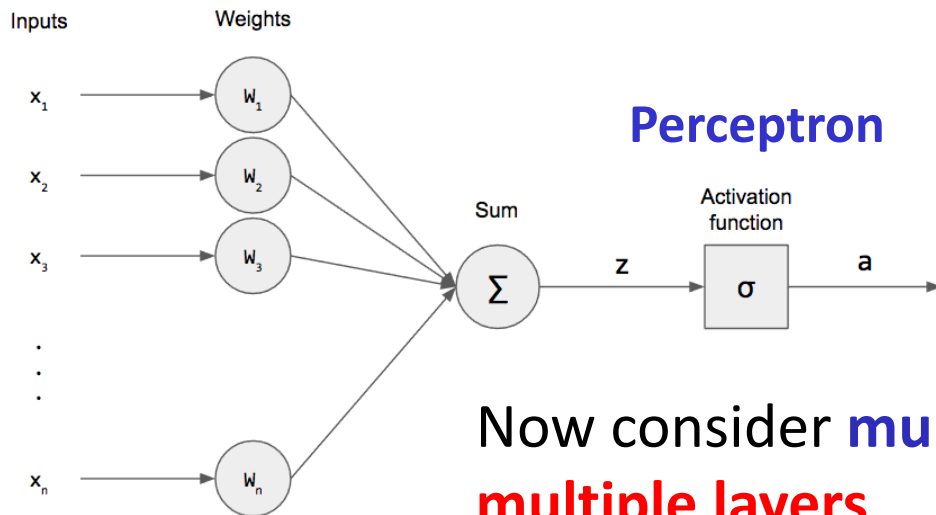
Biological Motivation

- The brain has 100 billion neurons, in parallel
- Mathematical model of neurons
- McCulloch and Pitts (1943) binary threshold model



If combined inputs exceed threshold, neuron **activates**, otherwise it remains **dormant**

Third Generation: ANNs, Deep Learning



$$\text{Output}_{t+1} = \sigma \left(\sum_{j=1}^J w_j X_{jt} - \mu \right)$$

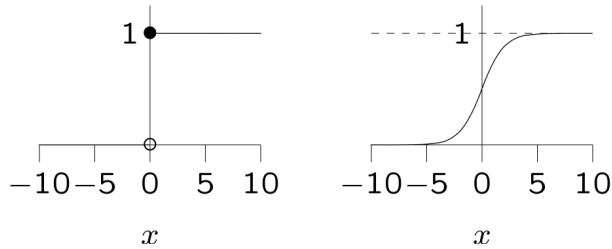
$$\sigma(u) = \begin{cases} 1 & \text{if } u \geq 0 \\ 0 & \text{if } u < 0 \end{cases}$$

Now consider **multiple perceptrons** and **multiple layers**

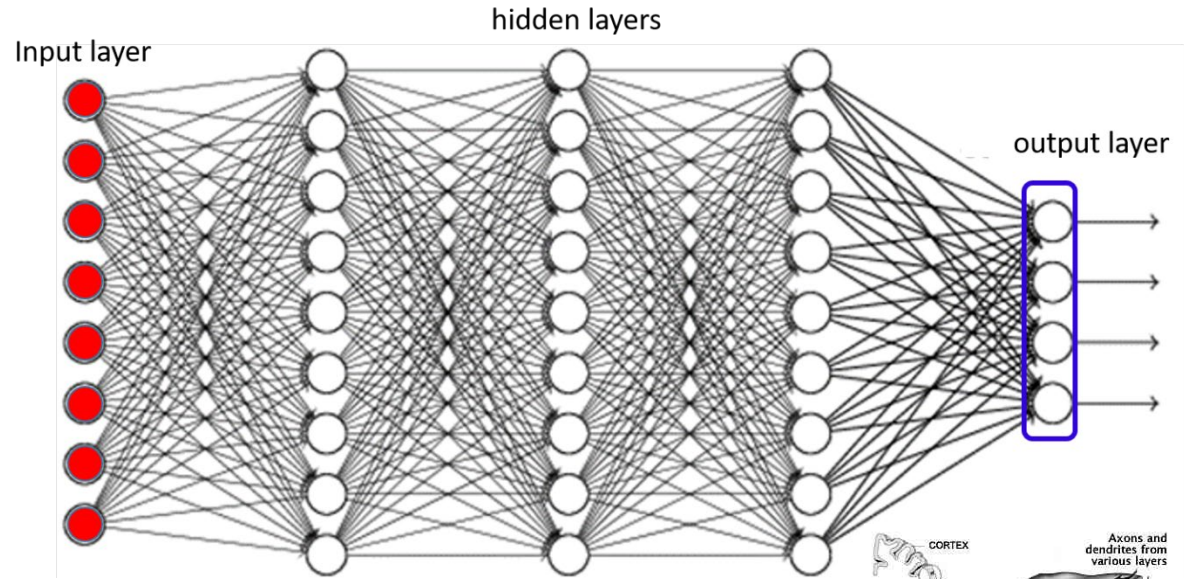
$$\sigma(w'_1 X) \quad \sigma(w'_2 X) \quad \dots \quad \sigma(w'_K X)$$

$$\hat{Y} = F \left(\sum_{k=1}^K \delta_k \sigma(w'_k X) \right)$$

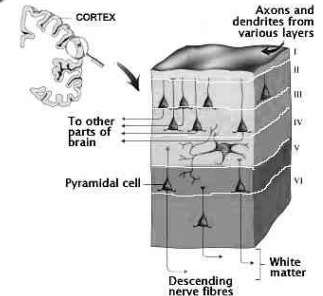
Third Generation: ANNs, Deep Learning



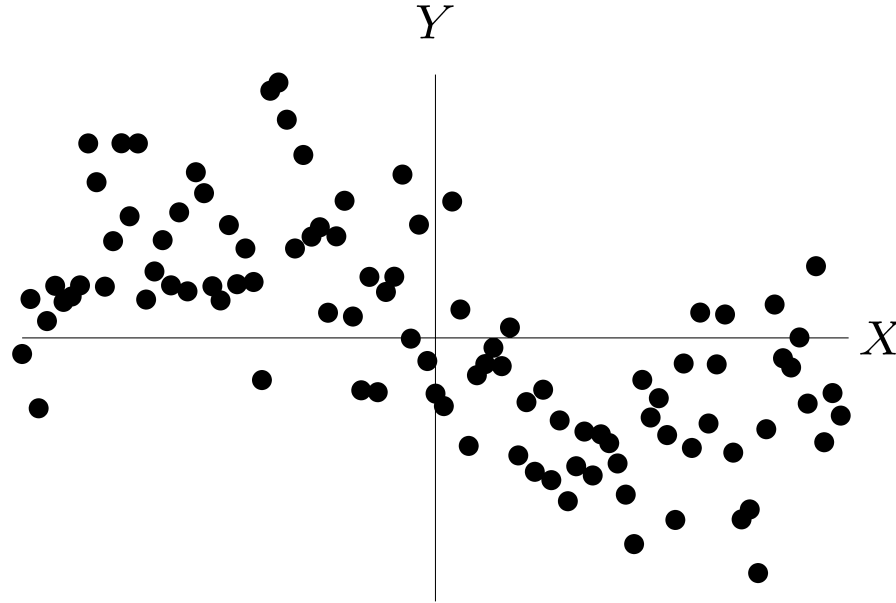
Activation Functions



“Deep Learning”



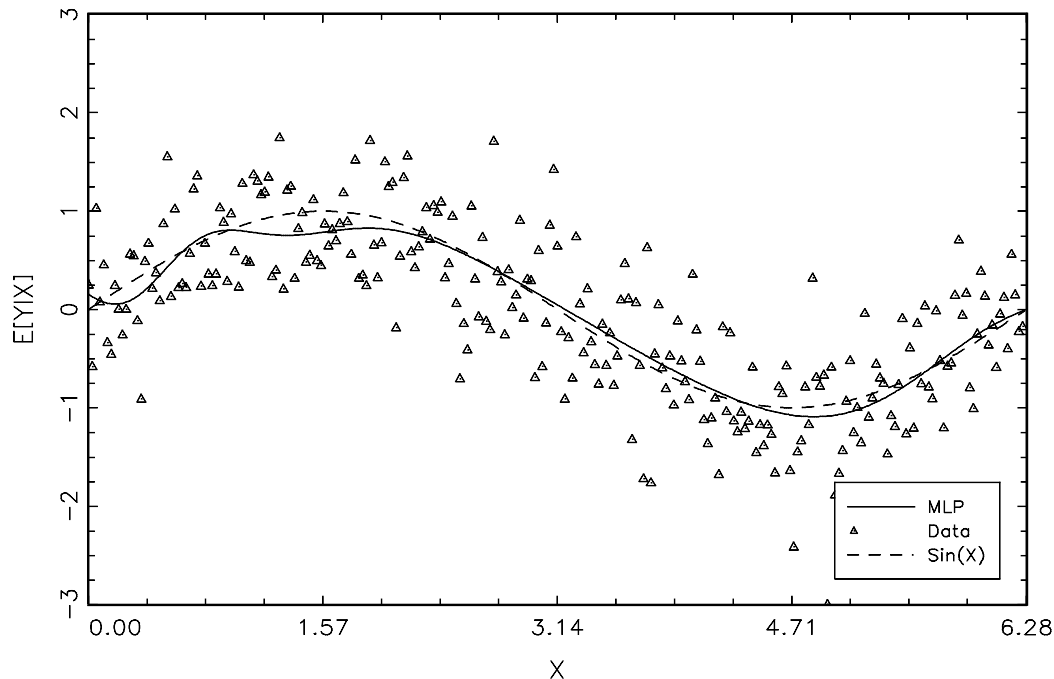
What Can Neural Networks Do?



$$Y = \sin(X) + 0.5\mathcal{N}(0, 1)$$

What Can Neural Networks Do?

MLP Estimator of $Y = \sin(X) + 0.5Z$



$$\begin{aligned}\hat{Y}_t = & 5.282 - 14.576 \cdot \sigma(-1.472 + 1.869X_t) \\ & - 5.411 \cdot \sigma(-2.628 + 0.641X_t) \\ & - 3.071 \cdot \sigma(13.288 - 2.347X_t) \\ & + 6.320 \cdot \sigma(-2.009 + 4.009X_t) \\ & + 7.892 \cdot \sigma(-3.316 + 2.484X_t)\end{aligned}$$

$$\sigma(u) = \frac{1}{1 + e^{-u}}$$

- Looks nothing like “sin(x)”
- Doesn't have to; is it a good approximation?

What Can Neural Networks Do?

- Very compact and easily parallelizable code:

```
A = 1 / (1 + EXP(-(X*GAMMA))) ;
```

```
B = DELTA0 + A*DELTA1;
```

```
C = Y - B ; SUMSQ = C'C;
```

- Computationally more efficient in some cases
- Depends on specific applications

Consider Derivative Pricing Models

- What if Black, Scholes, and Merton were never born?
- Could neural networks “discover” the formula?

Neural Networks and Derivatives Pricing

THE JOURNAL OF FINANCE • VOL. XLIX, NO. 3 • JULY 1994

1994

A Nonparametric Approach to Pricing and Hedging Derivative Securities Via Learning Networks

JAMES M. HUTCHINSON, ANDREW W. LO, and
TOMASO POGGIO*



- Simulate Black-Scholes/Merton prices and train NNs
- Can NNs “learn” Black-Scholes/Merton?
- Answer: **YES!**

Neural Networks and Derivatives Pricing

$$\begin{aligned}
 \widehat{C/X} = & -0.06 \sqrt{\left[\begin{matrix} S/X - 1.35 \\ T - t - 0.45 \end{matrix} \right]' \begin{bmatrix} 59.79 & -0.03 \\ -0.03 & 10.24 \end{bmatrix} \begin{bmatrix} S/X - 1.35 \\ T - t - 0.45 \end{bmatrix} + 2.55} \\
 & - 0.03 \sqrt{\left[\begin{matrix} S/X - 1.18 \\ T - t - 0.24 \end{matrix} \right]' \begin{bmatrix} 59.79 & -0.03 \\ -0.03 & 10.24 \end{bmatrix} \begin{bmatrix} S/X - 1.18 \\ T - t - 0.24 \end{bmatrix} + 1.97} \\
 & + 0.03 \sqrt{\left[\begin{matrix} S/X - 0.98 \\ T - t + 0.20 \end{matrix} \right]' \begin{bmatrix} 59.79 & -0.03 \\ -0.03 & 10.24 \end{bmatrix} \begin{bmatrix} S/X - 0.98 \\ T - t + 0.20 \end{bmatrix} + 0.00} \\
 & + 0.10 \sqrt{\left[\begin{matrix} S/X - 1.05 \\ T - t + 0.10 \end{matrix} \right]' \begin{bmatrix} 59.79 & -0.03 \\ -0.03 & 10.24 \end{bmatrix} \begin{bmatrix} S/X - 1.05 \\ T - t + 0.10 \end{bmatrix} + 1.62} \\
 & + 0.14S/X - 0.24(T-t) - 0.01
 \end{aligned}$$

Vs.

$$C(S, t) = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$$

$$\begin{aligned}
 \text{where } d_1 & \equiv \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} \\
 d_2 & \equiv \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}
 \end{aligned}$$

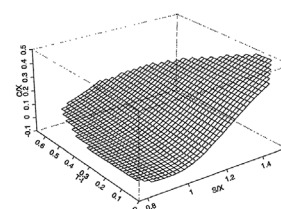
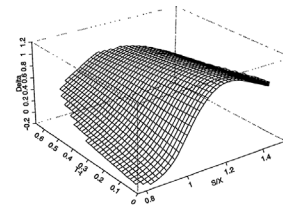
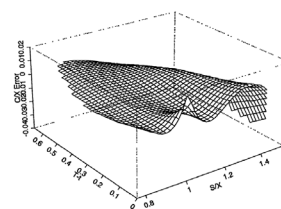
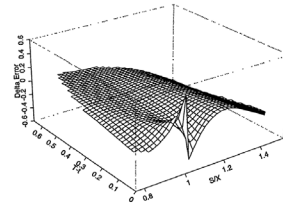
Neural Networks and Derivatives Pricing

Table XVI

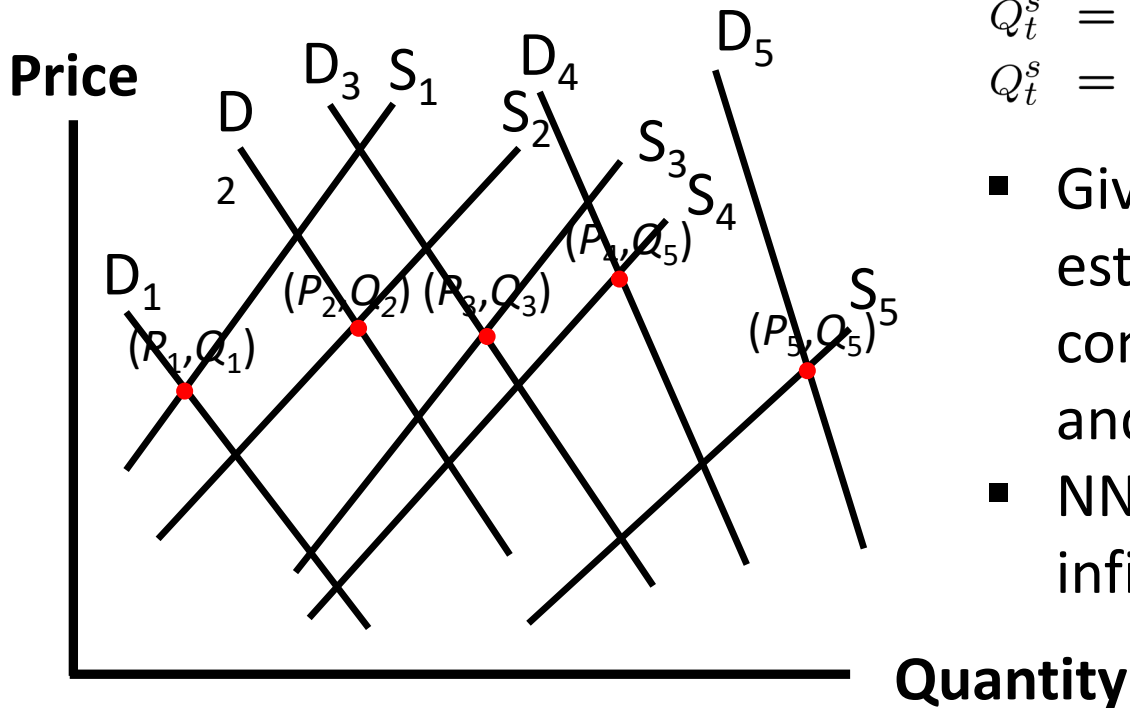
Delta-Hedging Prediction Error for the Out-of-Sample S&P 500 Data from July 1988 to December 1991

Data excludes the subperiods directly influenced by the October 1987 crash, averaged across all training/test sets. RBF indicates radial basis function; PPR indicates projection pursuit regression; MLP indicates multilayer perceptrons; and B-S indicates Black-Scholes formula.

	Linear-1	Linear-2	RBF	PPR	MLP	B-S	$\bar{C}(0)$
Short term							
In the money	6.70	4.92	5.04	4.52	4.94	4.42	24.26
Near the money	8.70	4.12	3.49	3.37	3.42	2.76	8.04
Out of the money	8.38	2.71	2.17	2.21	1.63	1.59	1.00
Medium term							
In the money	9.48	6.41	6.70	6.53	5.62	5.93	35.88
Near the money	8.82	6.93	4.18	5.02	4.54	5.31	10.62
Out of the money	11.27	4.69	2.53	2.73	2.32	2.55	2.74
Long term							
In the money	8.23	6.14	7.24	11.40	5.60	7.58	35.27
Near the money	8.55	8.58	6.37	5.55	5.17	6.18	16.04
Out of the money	12.13	7.35	3.54	5.29	4.36	5.02	5.86

(a) Network call price $\widehat{C}/X$ (b) Network delta $\widehat{\frac{\partial C}{\partial S}}$ (c) Call price error $\widehat{C}/X - C/X$ (d) Delta error $\widehat{\frac{\partial C}{\partial S}} - \frac{\partial C}{\partial S}$

What Can't Neural Networks Do (Yet)?



$$Q_t^d = d_0 + d_1 P_t + d_2 I_t + \epsilon_t^d$$

$$Q_t^s = s_0 + s_1 P_t + s_2 C_t + \epsilon_t^s$$

$$Q_t^s = Q_t^d = Q_t \quad \text{Equilibrium}$$

- Given inputs $\{(P_t, Q_t, I_t, C_t)\}$, estimate impact of 10% consumption tax on prices and quantities
- NNs can't do this, even with infinite amount of data

What Can't Neural Networks Do (Yet)?

Lancet Digital Health 2022

Articles

Combining the strengths of radiologists and AI for breast cancer screening: a retrospective analysis



Christian Lebig*, Moritz Brehmer*, Stefan Bunk, Danalyn Byng, Katja Pinkert, Lale Umutlu†



Summary

Background We propose a decision-referral approach for integrating artificial intelligence (AI) into the breast-cancer screening pathway, whereby the algorithm makes predictions on the basis of its quantification of uncertainty. Algorithmic assessments with high certainty are done automatically, whereas assessments with lower certainty are referred to the radiologist. This two-part AI system can triage normal mammography exams and provide post-hoc

Lancet Digit Health 2022;
4: e507–19

See [Comment](#) page e478

*Joint first authors

Fourth Generation: LLMs

2020's: Transforms and Foundation Models

nar·ra·tive \ 'ner-ə-tiv\ *noun*

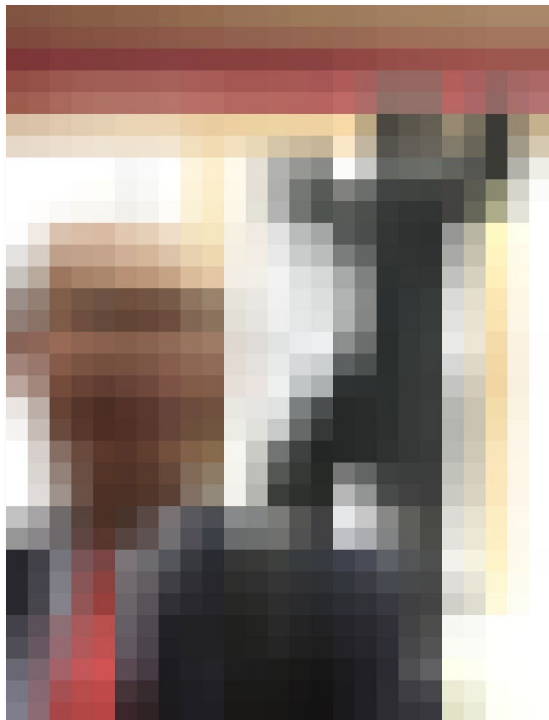
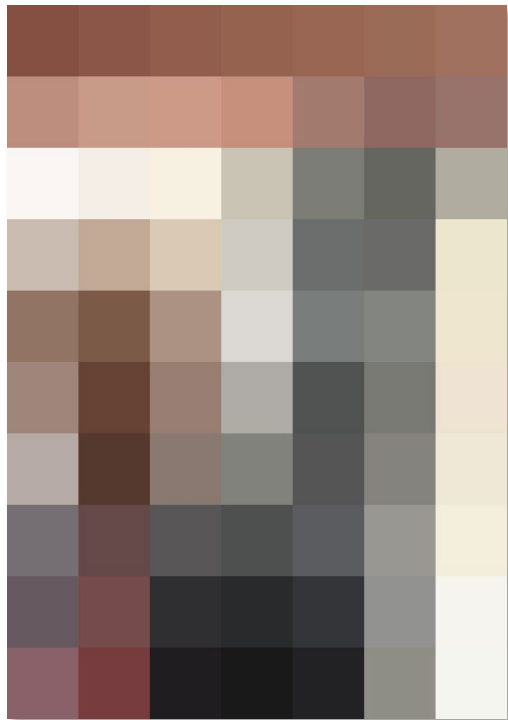
something that is narrated: **story, account**

More Relevant Definition for AGI:

- A **sequence** of **facts** that are a **causal explanation and a recipe**
- Provides control over a potentially hostile environment, yielding an evolutionary advantage

Narrative

Friend or Foe?



Friend or Foe?

- Gender and sex orientation (4)
- Marital status (3)
- Race/ethnicity (4)
- Age group (4)
- Current home state (50)
- Religious affiliation (4)
- Political party (3)
- Economic status (3)
- Education (3)

José

GM
Single
Latino
Young Prof.
CA
None
Democrat
Mid. Class
MBA

Helen

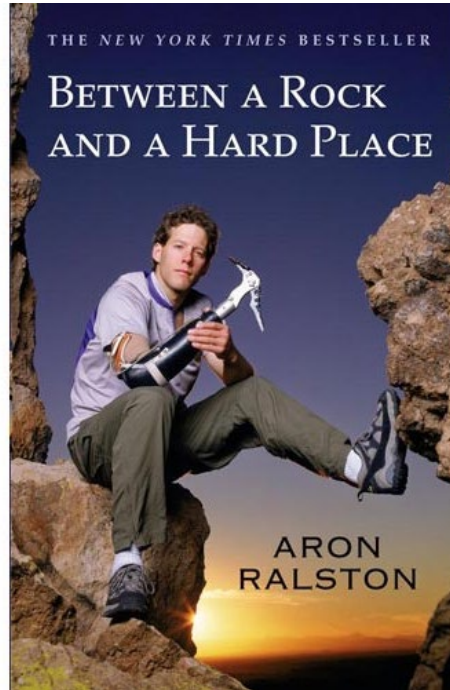
HF
Married
White
Mid. Age
TX
Christian
Republican
Affluent
BA

1,036,800 Possible Types!

But Beware of Learning With Sparse Data

The Power of Narrative

Aron Lee Ralston, 4/26/03



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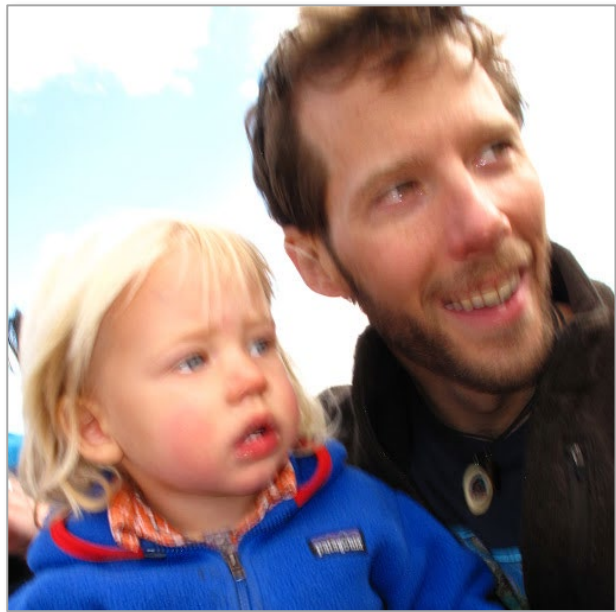
The Power of Narrative

A blond three-year-old boy in a red polo shirt comes running across a sunlit hardwood floor in what I somehow know is my future home. By the same intuitive perception, I know the boy is my own. I bend to scoop him into my left arm, using my handless right arm to balance him, and we laugh together as I swing him up to my shoulder... Then, with a shock, the vision blinks out. I'm back in the canyon, echoes of his joyful sounds resonating in my mind, creating a subconscious reassurance that somehow I will survive this entrapment. Despite having already come to accept that I will die where I stand before help arrives, now I believe I will live.

That belief, that boy, changes everything for me.

– Aron Lee Ralston (2005)

The Power of Narrative



**What Can GenAI Do With
Its Abilities To Generate
And Process Narrative??**

More tomorrow...

Thank You!